POLITECNICO
MILANO 1863

System Identification of Multirotor UAV Longitudinal Dynamic



Mohamed Ghoraba

Karim Ahmed Mohamed Doubie

Riccardo Cislaghi

Estimation and Learning in Aerospace

Professor: Marco Lovera

Project Overview

- **Task 1 - Grey Box Identification**

  Identification of the **longitudinal dynamics** using a grey-box model.

  → Parameter estimation, uncertainty analysis, validation with a 3211 sequence

- **Task 2 – Sensitivty Analysis**

  Analysis of the identification result with respect to perturbation in measurement noise proprietes

  → Repeat identification for different sensor noise trade-offs and select the best model.

- **Task 3 - Monte Carlo Analysis**

  Statistical Analysis of the robustness of the identification process for the best model

  → Stathistical analysis of parameters, poles/zeros and frequencies response functions

Linearized Model for Longitudinal Dynamics

State equation

$$\underbrace{\begin{bmatrix}\dot{u}\\ \dot{q}\\ \dot{\theta}\end{bmatrix}}_{\dot{x}} = \underbrace{\begin{bmatrix}X_u & X_q & -g\\ M_u & M_q & 0\\ 0 & 1 & 0\end{bmatrix}}_{A(\Theta)} \underbrace{\begin{bmatrix}u\\ q\\ \theta\end{bmatrix}}_{x} + \underbrace{\begin{bmatrix}X_\delta\\ M_\delta\\ 0\end{bmatrix}}_{B(\Theta)} \underbrace{\delta_{lon}}_{u}$$

$$\Theta = \begin{bmatrix}X_u\\ X_q\\ M_u\\ M_q\\ X_\delta\\ M_\delta\end{bmatrix}$$

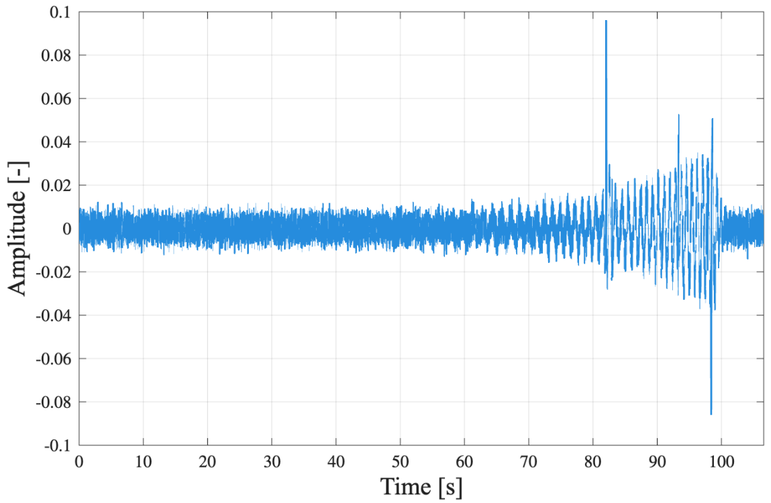
Output equation

$$\underbrace{\begin{bmatrix}q\\ a_x\end{bmatrix}}_{y} = \underbrace{\begin{bmatrix}0 & 1 & 0\\ X_u & X_q & 0\end{bmatrix}}_{C(\Theta)} \underbrace{\begin{bmatrix}u\\ q\\ \phi\end{bmatrix}}_{x} + \underbrace{\begin{bmatrix}0\\ X_\delta\end{bmatrix}}_{D(\Theta)} \underbrace{\delta_{\text{lon}}}_{u}$$

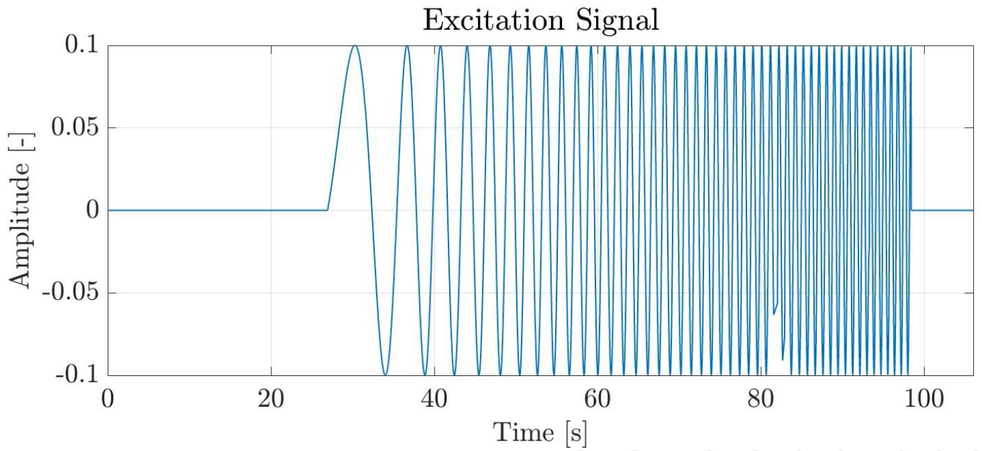
u	Longitudinal (body) velocity [m/s]
q	Pitch rate [rad/s]
θ	Pitch angle [rad]
δ_{lon}	Pitch moment [normalised to -1, $+1$ range]

INPUT/OUTPUT DATA

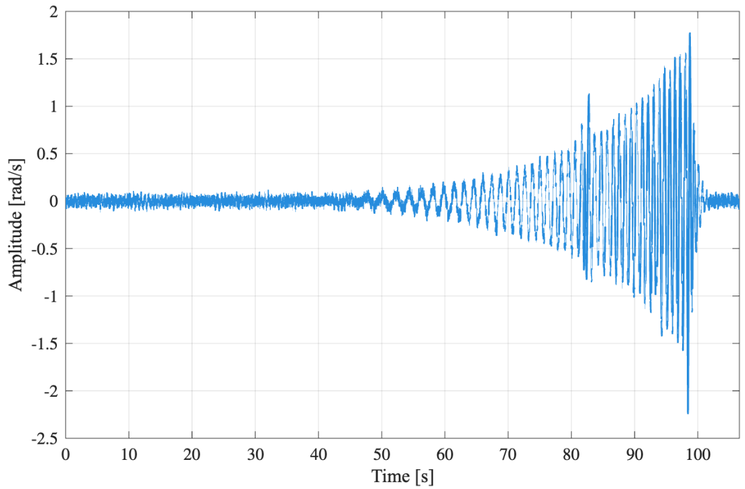
INPUT :Normalised Pitching Moment

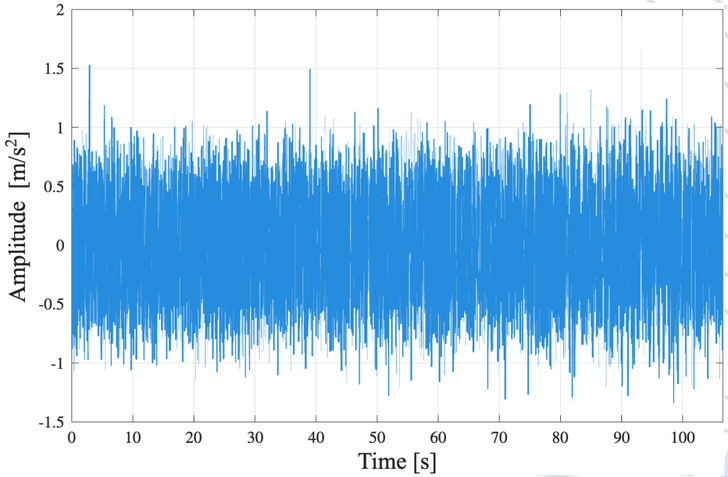


Excitation Signal



OUTPUT : Pitch rate & Longudinal acceleration





Task 1 :

Grey Box model identification



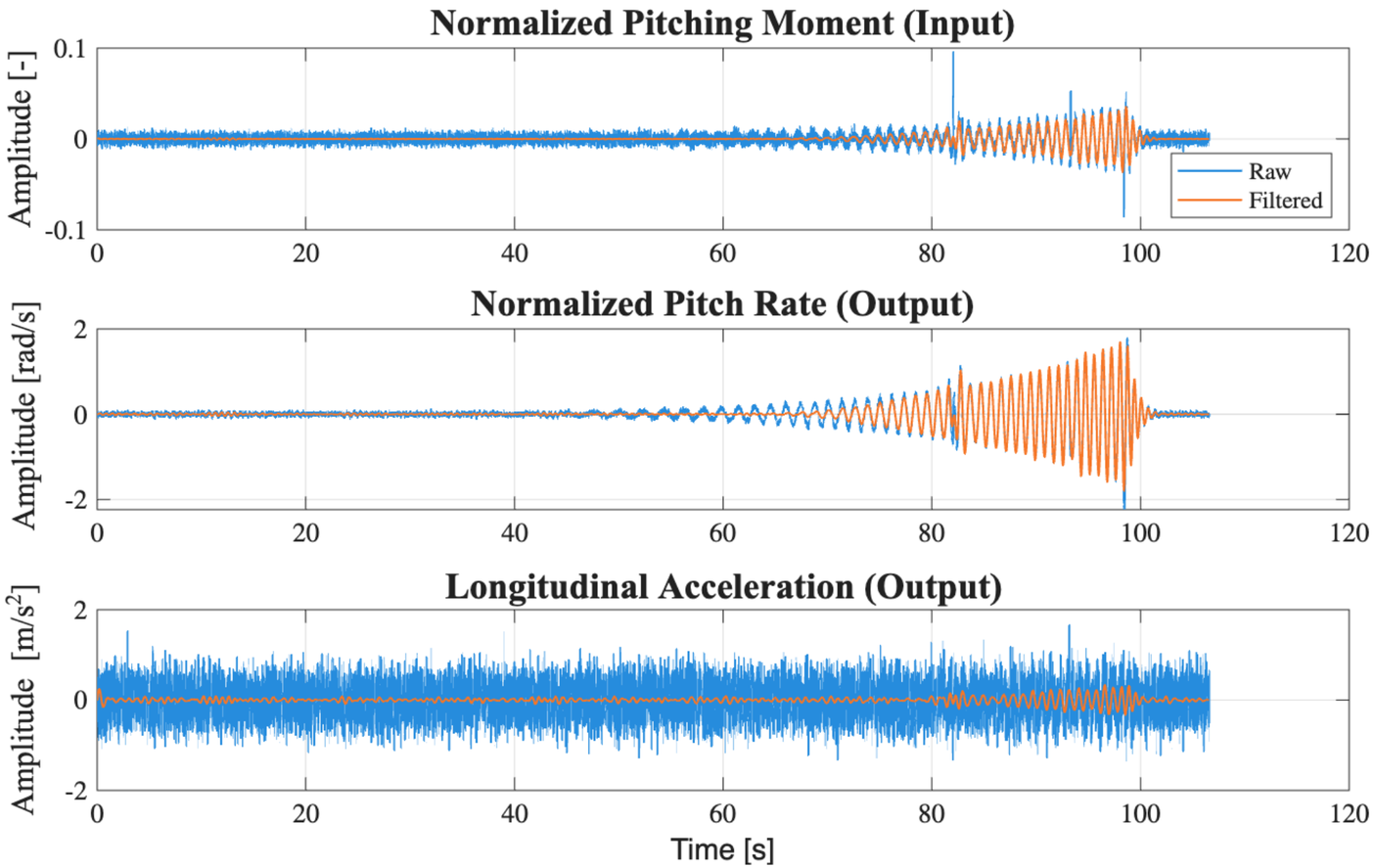
Preprocessing Pipeline

To improve the quality of the estimated parameters and guarantee convergence:

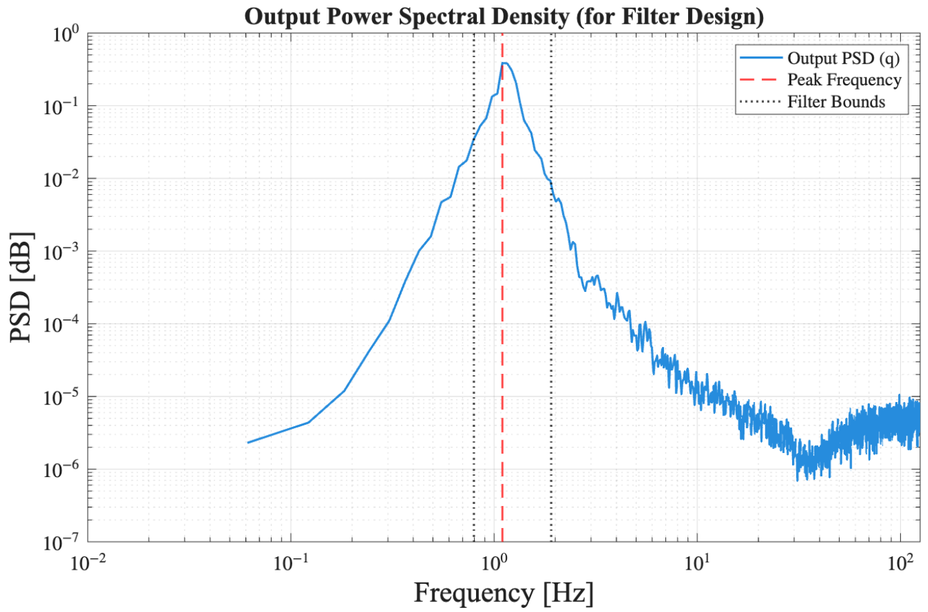
- **Data Prefiltering**
  - → remove offsets and apply a band-pass filter around the PSD peak
- **FRF Estimation (Welch Method)***
  - → obtain a non-parametric frequency response from filtered I/O data with coherence check
- **Model Identification**
  - → Method 1: apply *greyest* on the preprocessed FRF data
  - → Method 2: Parametric Identification (Output-Error Method)
- **Result Validation**
  - → assess the frequency range and refine the identified model

Data Prefiltering (For OE Method only)

$$\{u(t_k), y(t_k)\} \longrightarrow \{u_f(t_k), y_f(t_k)\}$$



Power Spectral Density



A 4th-order Butterworth **band-pass filter** is applied around the ***ω* peak** frequency of the estimated output PSD

Method 1: Non-Parametric FRF

We need the FRF from I/O data to run greyest

Option A

Direct FFT on I/O data

- Build `iddata` --> compute FFT
- Perform system identification

Option B

Welch Method

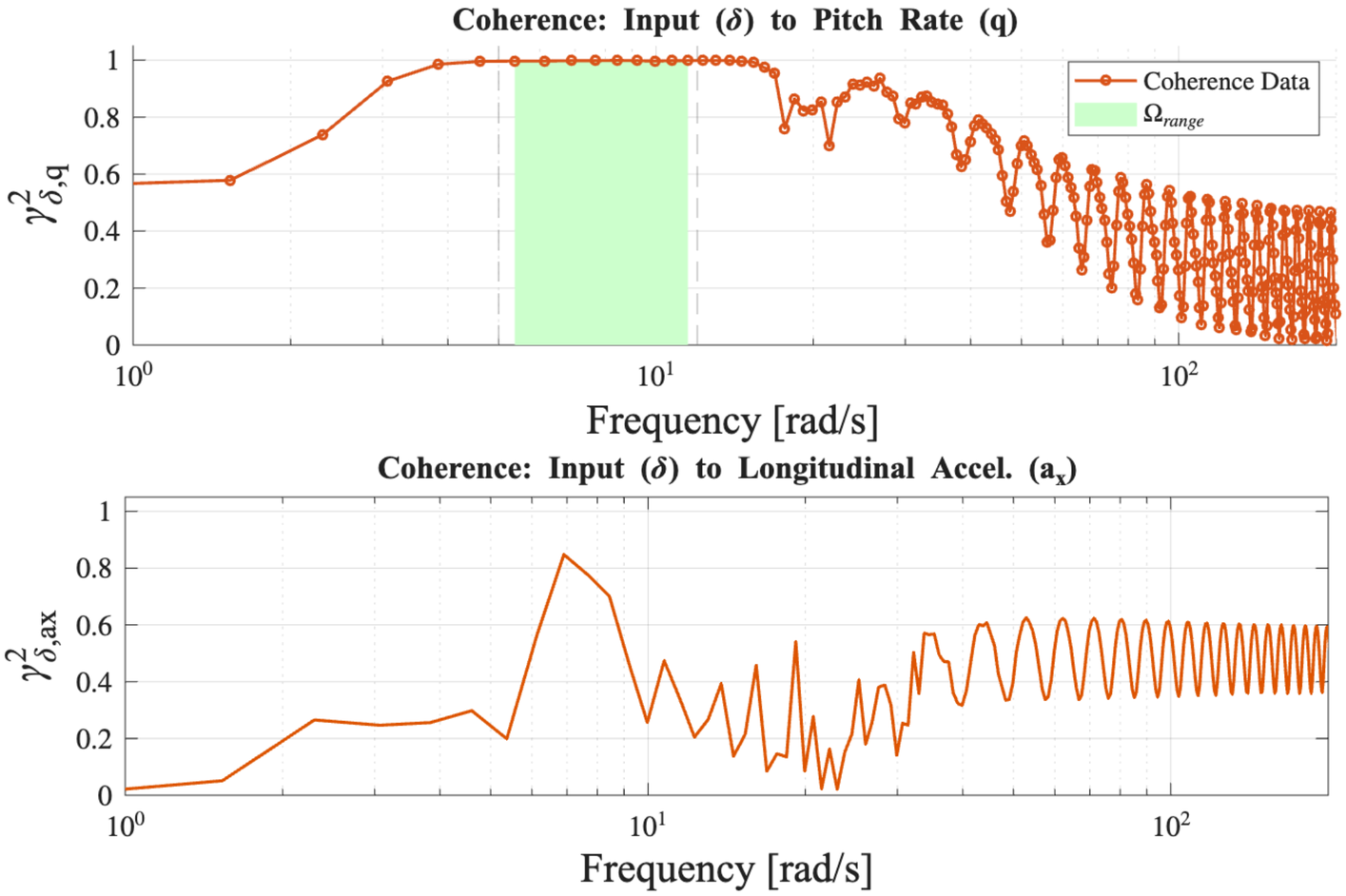
- Filter I/O data → average + window
- Reduce variances thanks to smoothing
- In closed-loop → used unbiased estimator formulation

$$\hat{S}(j\omega) = \frac{\hat{\Phi}_{u\delta}(j\omega)}{\hat{\Phi}_{\delta}(j\omega)} \qquad \hat{SG}(j\omega) = \frac{\hat{\Phi}_{y\delta}(j\omega)}{\hat{\Phi}_{\delta}(j\omega)}$$

$$\hat{G}_m(j\omega) = \frac{\hat{SG}(j\omega)}{\hat{S}(j\omega)}$$

Coherence Function

Coherence function highlights the frequencies where the input excites the system properly and the output is reliable.
It is used to **validate frequency ranges** and to ensures that non parametric FRF estimates are based only on **meaningful data.**



$$\hat{\gamma}^2_{\delta,\,y}(j\omega) = \frac{\left|\hat{\Phi}_{\delta,\,y}(j\omega)\right|^2}{\hat{\Phi}_{\delta}(j\omega)\hat{\Phi}_{y}(j\omega)}$$

$\hat{\Phi}_{\delta}(j\omega)$ → input power spectral density (pwelch)

$\hat{\Phi}_{y}(j\omega)$ → output power spectral density (pwelch)

$\hat{\Phi}_{\delta,\,y}(j\omega)$ → cross-power spectral density between input and output (cpsd)

Frequency range: 0.76 Hz<f< 1.85 Hz

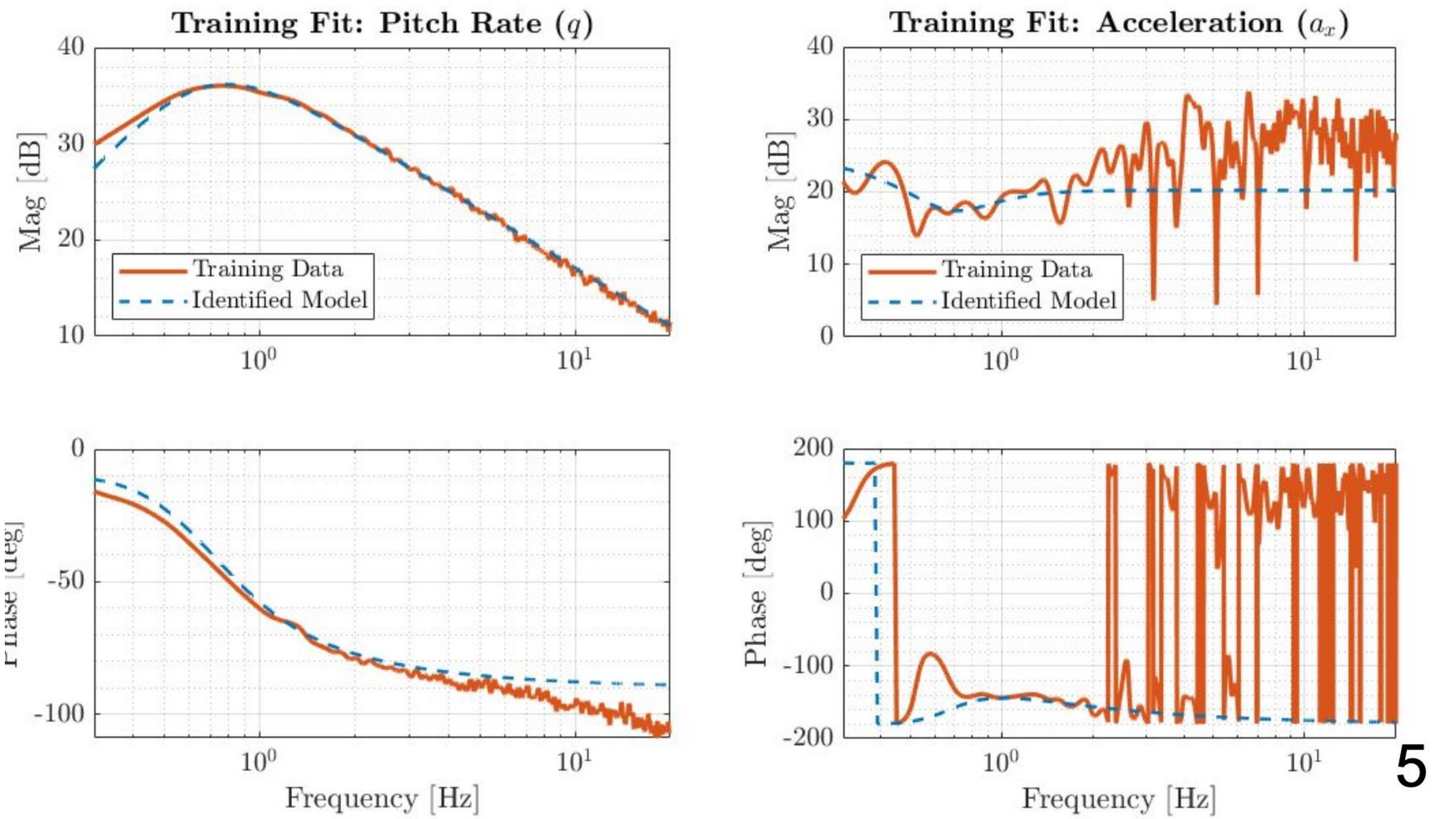
Method 1 : Identification results (Option A)

Estimated Model

Parameter	Value	Uncertainty (%)
X_u	-0.1324	60.56
X_q	0.1314	19.40
M_u	-5.9794	0.99
M_q	-2.6456	0.54
X_d	-10.2926	6.36
M_d	450.4785	0.13

Output	OE Model
Pitch Rate (q)	99.48%
Acceleration (a_x)	38.16%

Response of the Estimated Model



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Method 2: Parametric identification (OE Method)

Time Domain preprocessing:

- 1° order Butterworth low-pass filter (10 hz) to a_x
- Subtracted the mean from the total input and output

Freq Domain preprocessing:

- 1° Hanning window with 50% overlap to compute S_{uu}, S_{qq}, S_{aa}, S_{uq}, S_{ua_x}
- Computed non-parametric FRFs

$$\hat{G}q = Suq/S_{uu} \quad \hat{G}a_x = Sua_x/S_{uu}$$

- Used the same freq range from greyest method

Stacking

- Data is stacked

$$Y(\omega_k) = \begin{bmatrix} \Re\left(G_q(\omega_k)\right) \\ \Re\left(G_{a_x}(\omega_k)\right) \\ \Im\left(G_q(\omega_k)\right) \\ \Im\left(G_{a_x}(\omega_k)\right) \end{bmatrix}$$

$$e(\omega_k, \theta) = Y_m(\omega_k) - Y(\omega_k, \theta)$$

Cost function $J(\theta) = \frac{1}{2}\sum_{k=1}^{K} e_k^T R^{-1} e_k$

Where R is the measurement error matrix, initialized with I_4 and updated at each iteration with

$$R \approx \frac{1}{K - n_\theta}\sum \left| e_k e_k^T \right|$$

Parameter Estimation

$$\boldsymbol{\theta_{new} = \theta - H^{-1}G}$$

Estimated Covariance

$$\boldsymbol{cov(\theta) = H^{-1}}$$

Hessian $H = \sum_{k=1}^{K} \left(\frac{\partial Y}{\partial \theta}\right)^T R^{-1} \left(\frac{\partial Y}{\partial \theta}\right)$

Gradient $G = \sum_{k=1}^{K} -e_k^T R^{-1} \left(\frac{\partial Y}{\partial \theta}\right)$

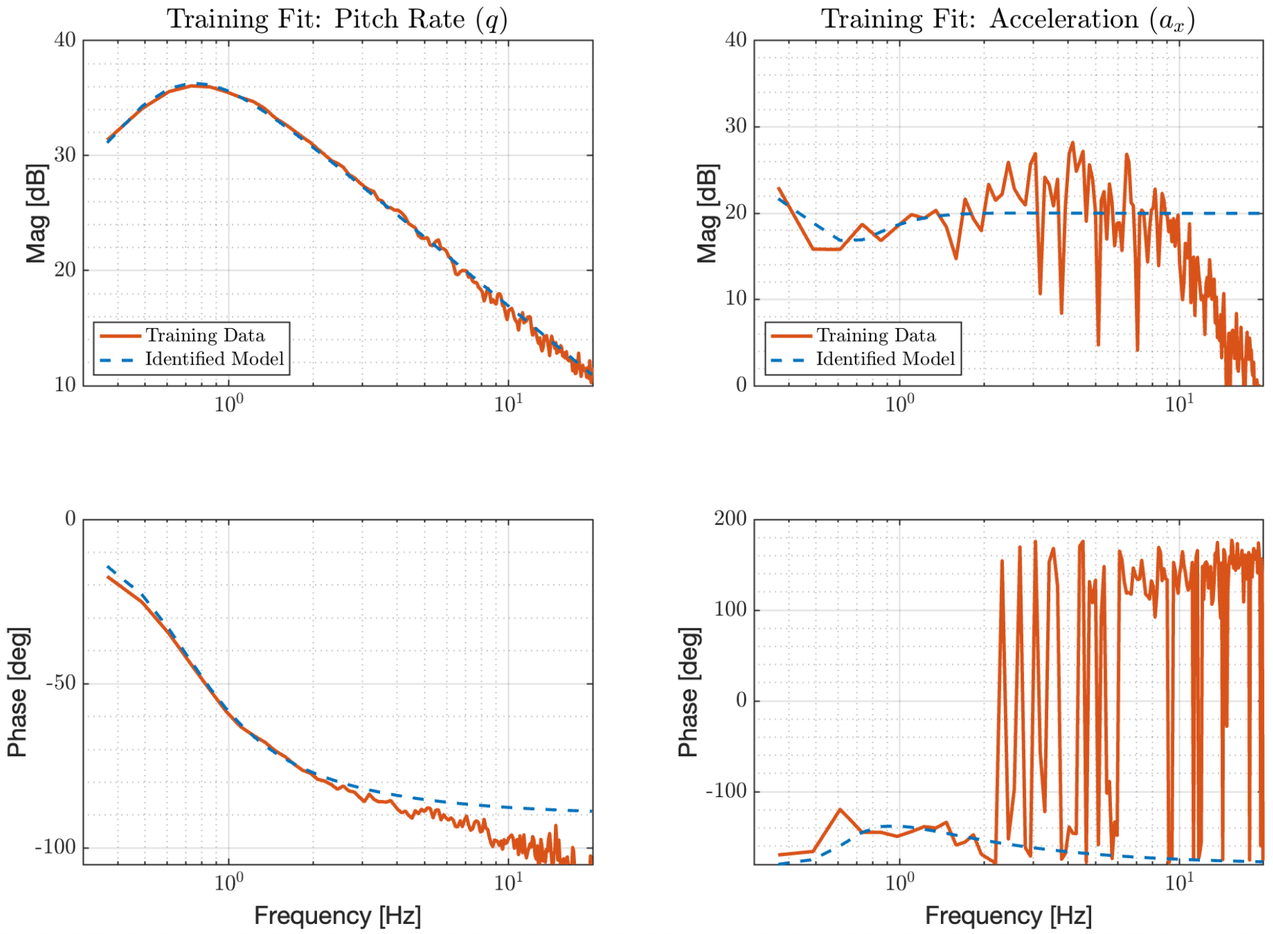
Method 2: Parametric identification (OE Method)

Estimated Model

Parameter	Value	Uncertainty (%)
X_u	-0.1079	52.38
X_q	0.1374	20.07
M_u	-5.0786	6.92
M_q	-2.6686	6.85
X_d	-9.9698	14.10
M_d	441.3740	1.76

Output	OE Model
Pitch Rate (q)	97.94%
Acceleration (a_x)	38.15%

Response of the Estimated Model



Asymptotic Variance

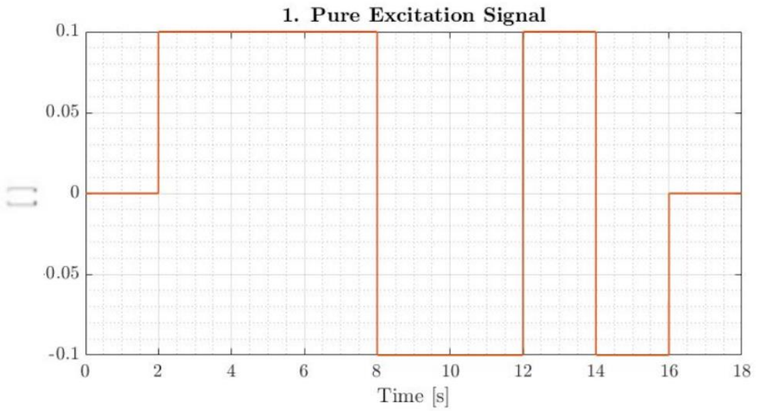
- The **asymptotic variance** quantifies the uncertainty of the parameter estimates in the large-sample limit; it is given by the inverse of the Fisher information matrix, with the diagonal entries of the covariance matrix corresponding to the variances of the individual parameters.

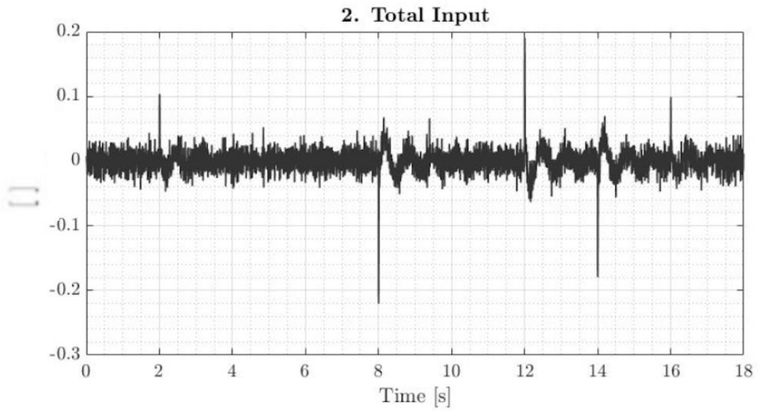
$$\mathrm{Cov}(\Theta) = \mathcal{M}^{-1} \qquad U_{\mathrm{rel}}(\%)_i = \frac{\sigma_{\hat{\theta}_i}}{\left|\hat{\theta}_i\right|} \times 100$$

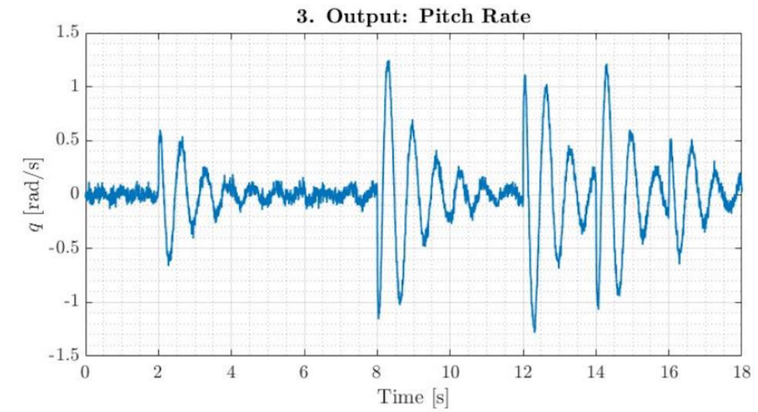
- When using **greyest**, these variances are found along the diagonal of the covariance matrix of the estimated parameters.

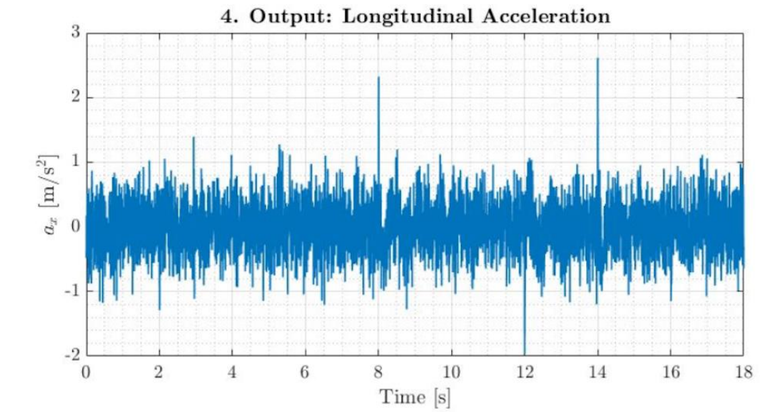
Parameter	Value	Asymptotic Var.
X_u	-0.1324	60.56%
X_q	0.1314	19.40%
M_u	-5.9794	0.99%
M_q	-2.6456	0.54%
X_d	-10.2926	6.36%
M_d	450.4785	0.13%

Model Validation via 3211 sequence



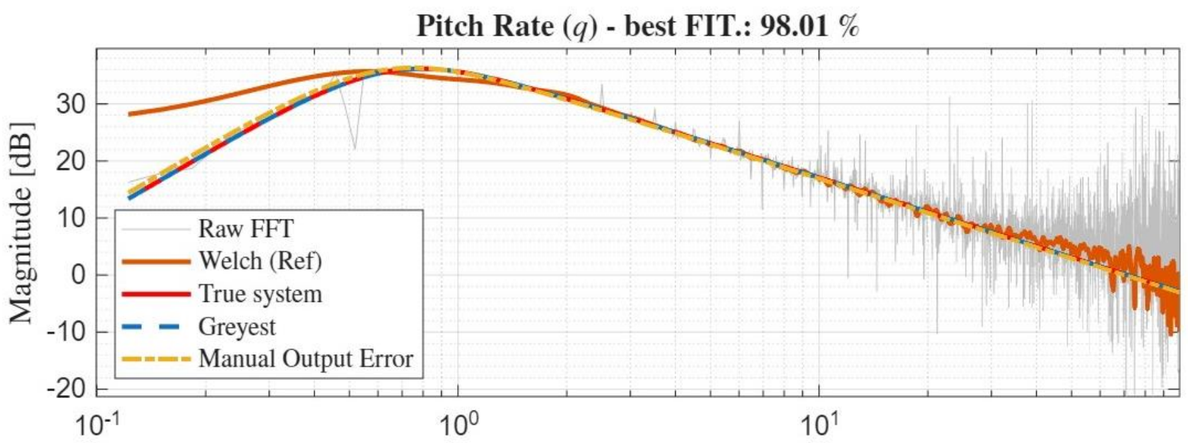


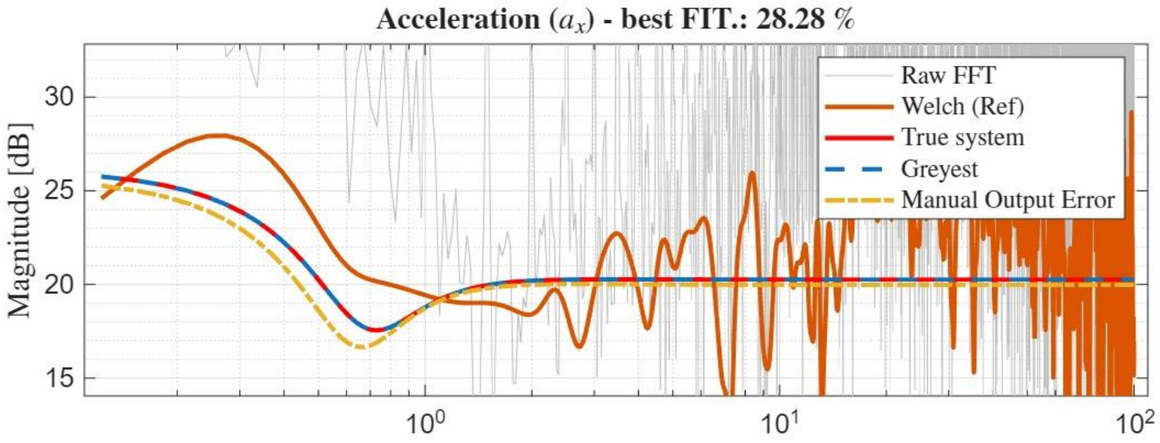


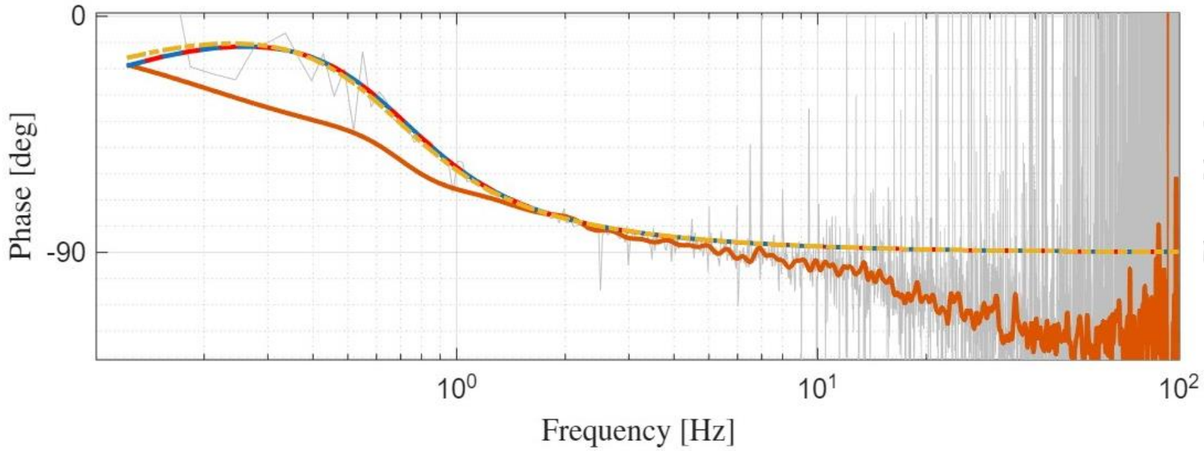


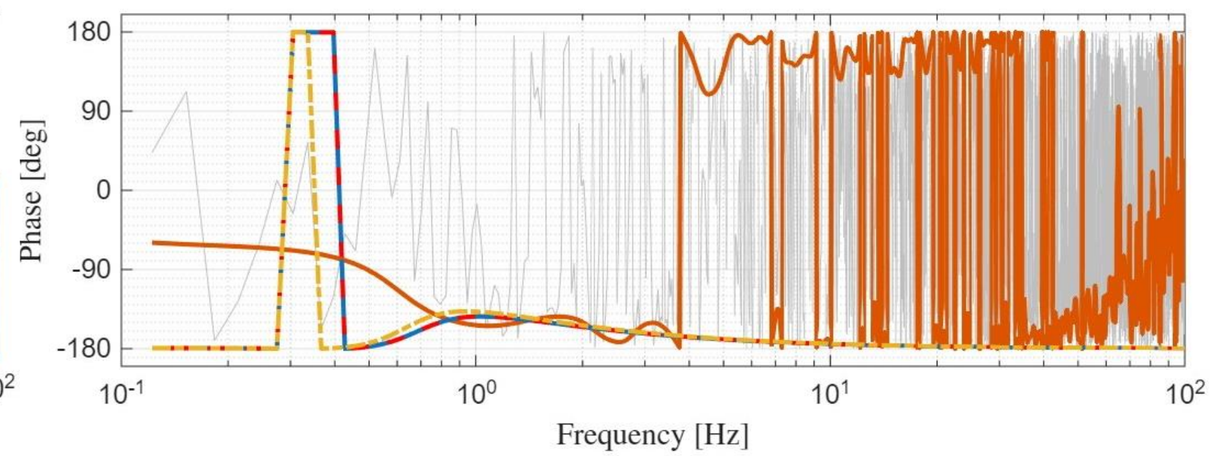
FIT	q	a_x
Greyest	98.01%	28.28%
Output Error	97.19%	28.53%

Model Validation via 3211 sequence









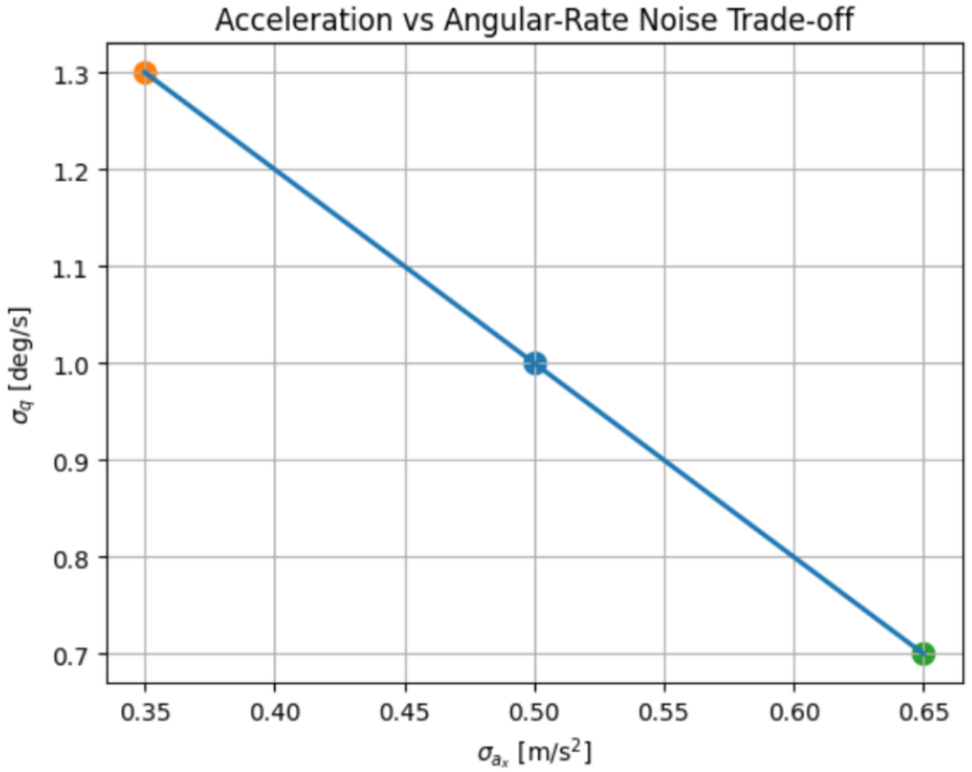
Task 2 :
Sensitivity Analysis



Sensitivity Analysis

Target: Sensitivity analysis to perturbation

Method: Repetition of the Task 1 process with different levels of noise in the outputs, simulating the accuracy of the sensors



Criteria

The fit for q for the 3 models showed little variation in fit percentages which indicated the q sensor noise had little effect on the model. However, the acceleration sensor noise level had signifant effect on the prediction of a_x, which **motivated using as criteria selecting the model which the highest a_x fit percentage**.

As expected, the model which the better a_x sensor had the highest fit percentage due to lower noise level.

Asymptotic Variance

$$U_{\text{rel}}(\%)_i = \frac{\sigma_{\hat{\theta}_i}}{\left|\hat{\theta}_i\right|} \times 100$$

Condition	Output	Base model	Better q sensor	Better a_x sensor
Identification	q	99.48%	99.60%	99.35%
	a_x	38.16%	28.78%	51.48%
Validation	q	98.01%	97.88%	98.07%
	a_x	28.28%	19.52%	42.08%

Best Model Performance

SELECTED MODEL

Parameter	Value	Asymptotic Var.
X_u	-0.1292	80.30%
X_q	0.1289	19.30%
M_u	-5.9827	1.34%
M_q	-2.6450	0.61%
X_d	-10.2457	5.86%
M_d	450.6015	0.10%

BASE MODEL

Parameter	Value	Asymptotic Var.
X_u	-0.1324	60.56%
X_q	0.1314	19.40%
M_u	-5.9794	0.99%
M_q	-2.6456	0.54%
X_d	-10.2926	6.36%
M_d	450.4785	0.13%

Asymptotic Variance

$$U_{\text{rel}}(\%)_i = \frac{\sigma_{\hat{\theta}_i}}{\left|\hat{\theta}_i\right|} \times 100$$

Output	Fit
q	99.35%
a_x	51.48%

Task 3 :
Monte Carlo Analysis



Monte Carlo analysis

The Monte Carlo Analysis has been carried out changing the seed structure inside the Simulink model, obtaining different realisations of the noise processes.

Specificely:

I. Dataset creation

$$\mathrm{DATASETS} = \left\{ \left(M_{\mathrm{tot}}^{(i)}, q^{(i)}, a_x^{(i)} \right) \right\}_{i=1,\ldots,N}$$

II. Model identification through Grey-box pipelines

$$\Theta_{T1}^{(i)} = \mathrm{greyBox}\left(\mathrm{DATASET}^{(i)} \right)$$

Monte Carlo analysis – Number of iterations

Number of Monte Carlo samples set to 1000

To choose coherently the number of iterations for the analysis, both theoretical and practical reasoning has been made.

Based on **detection guarantees**, we can detect failures that have a probability of happening epsilon equal to 0.5% with 99% confidence with a number of samples given by:

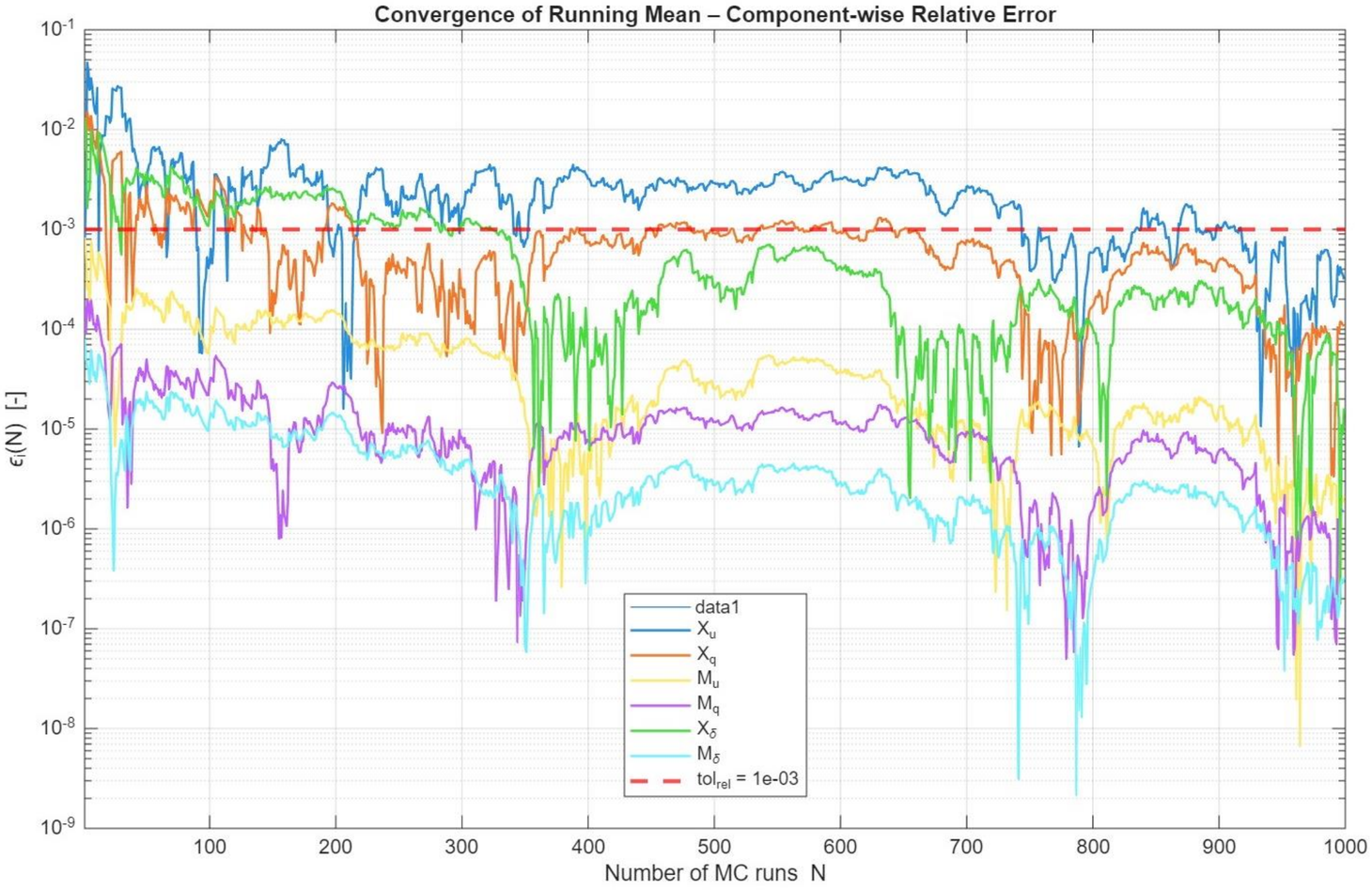
$$N > \frac{\log(1 - \text{confidence})}{\log(1 - \varepsilon)}$$

Monte Carlo analysis – Minimum number of iterations

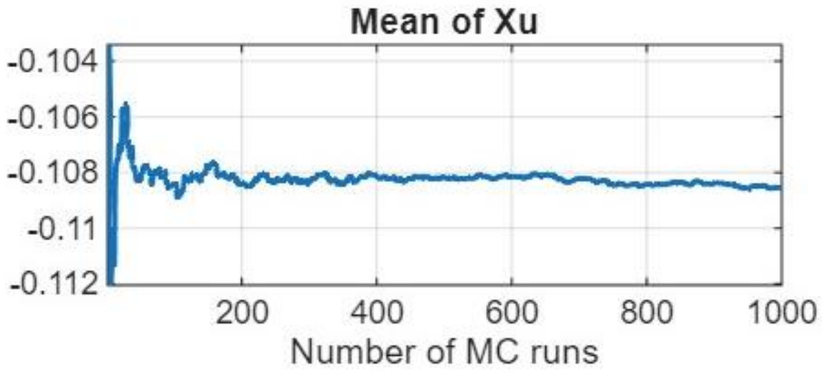
The number of simulations N is selected as the smallest value for which the running mean of all six estimated parameters simultaneously satisfies the convergence criterion against the full-ensemble reference $\bar{x}_i(N_{MC})$,with $N_{MC} = 1000$

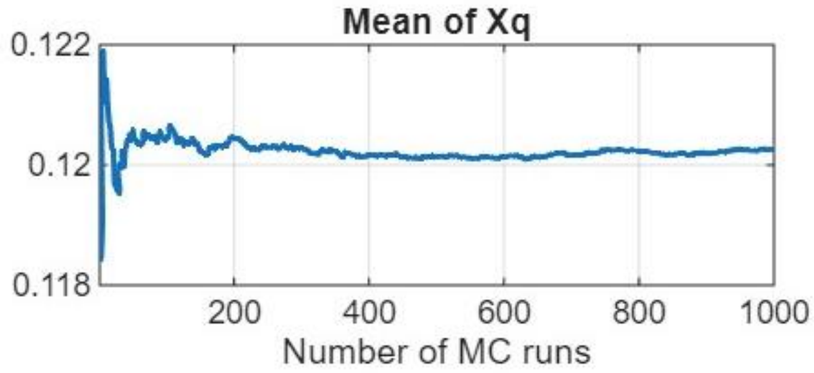
$$\epsilon_i(N) = \frac{|\bar{x}_i(N) - \bar{x}_i(N_{MC})|}{|\bar{x}_i(N_{MC})| + \epsilon} < tol_{rel} = 10^{-3}$$

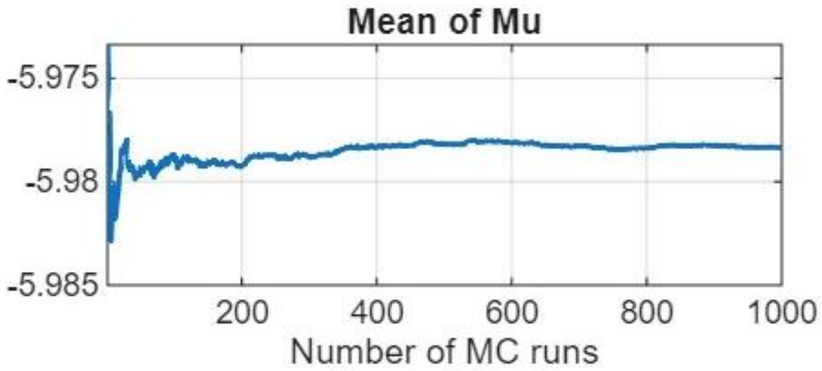
With $N_{MC} = 1000$ the error stays bound below the threshold

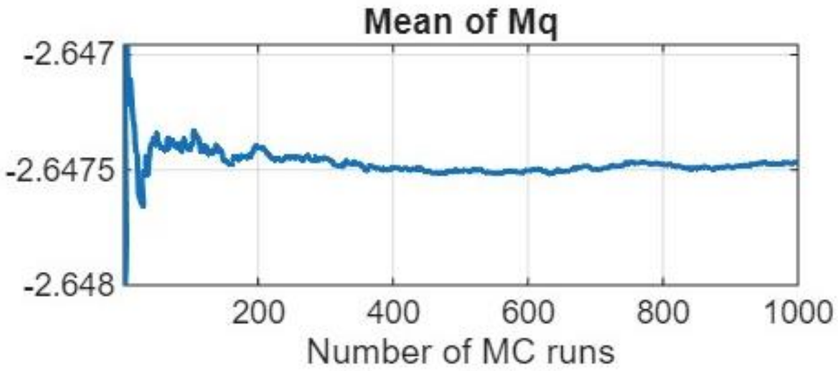


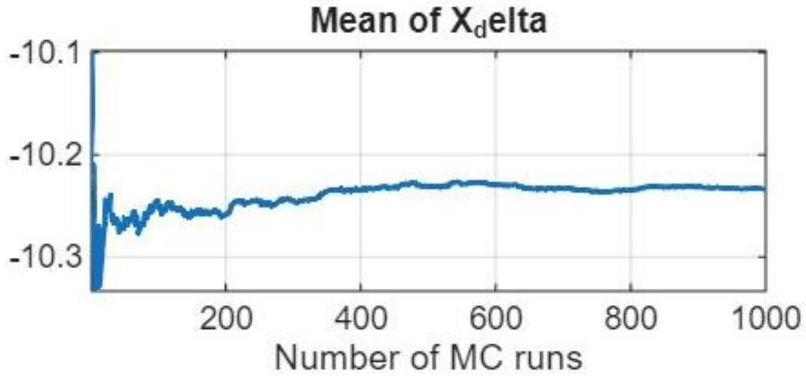
Monte Carlo analysis – Convergence of the mean

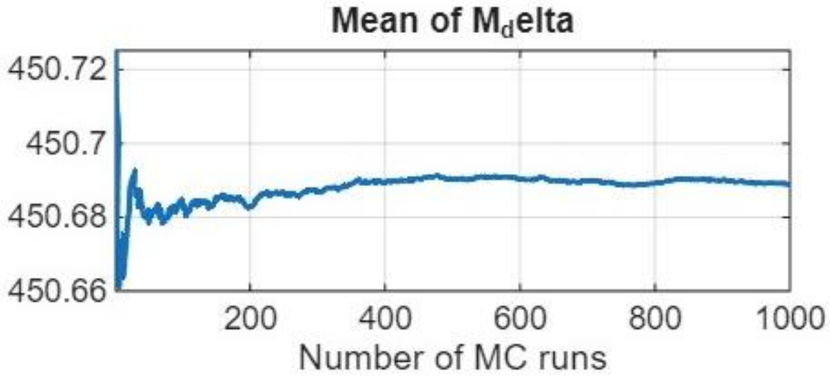






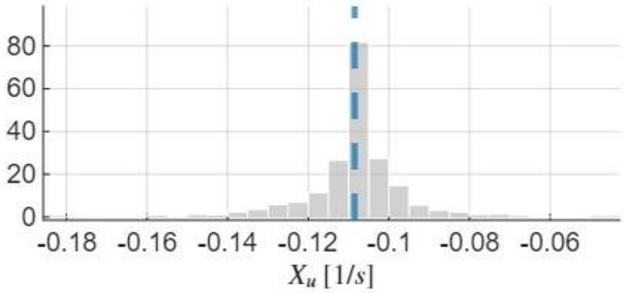


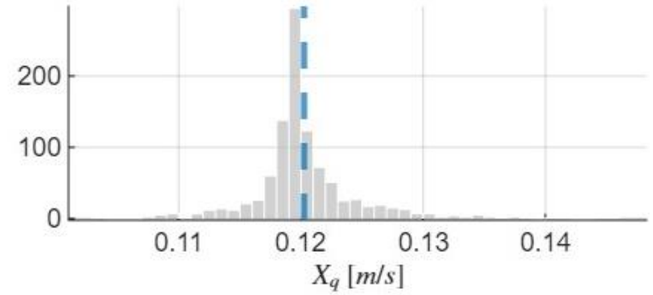


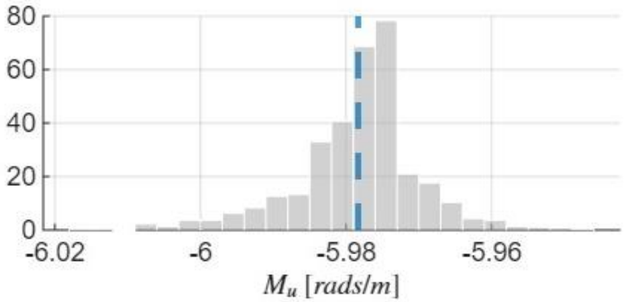


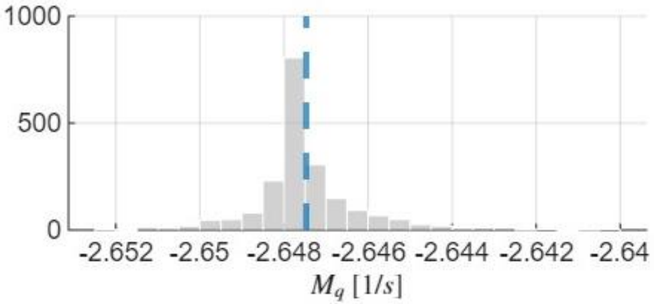
$$\hat{\boldsymbol{\theta}}_{mean} = \begin{bmatrix} \hat{X}_u \\ \hat{X}_q \\ \hat{M}_u \\ \hat{M}_q \\ \hat{X}_\delta \\ \hat{M}_\delta \end{bmatrix} = \begin{bmatrix} -0.1085 \\ 0.1203 \\ -5.9783 \\ -2.6475 \\ -10.2336 \\ 450.6891 \end{bmatrix}$$

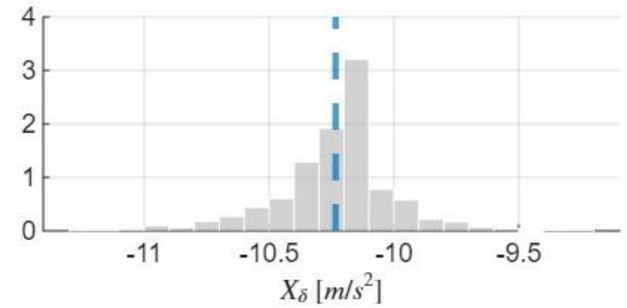
Monte Carlo analysis – Distribution of the Parameters

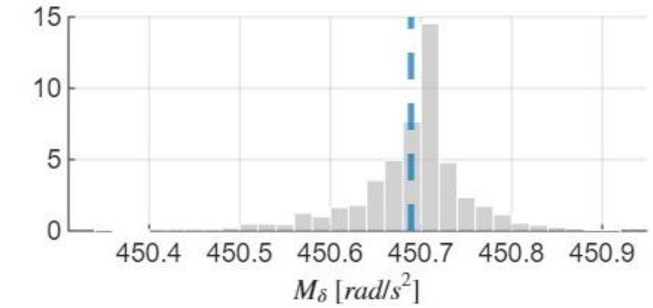












Mean

	Monte Carlo	True	% Error
X_u	-0.1085	-0.1068	1.59%
X_q	0.1203	0.1192	0.92%
M_u	-5.9783	-5.9755	0.05%
M_q	-2.6475	-2.6478	0.01%
X_δ	-10.2336	-10.1647	0.68%
M_δ	450.6891	450.71	<0.01%

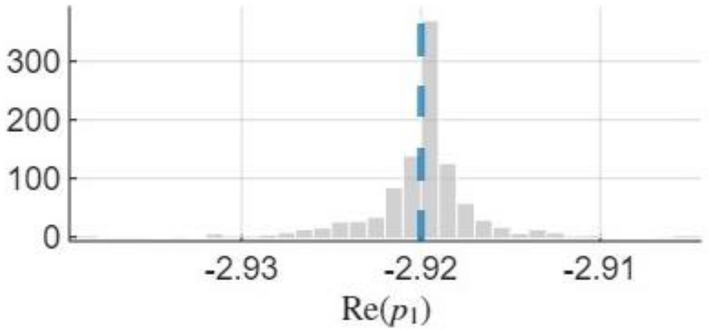
Standard deviation

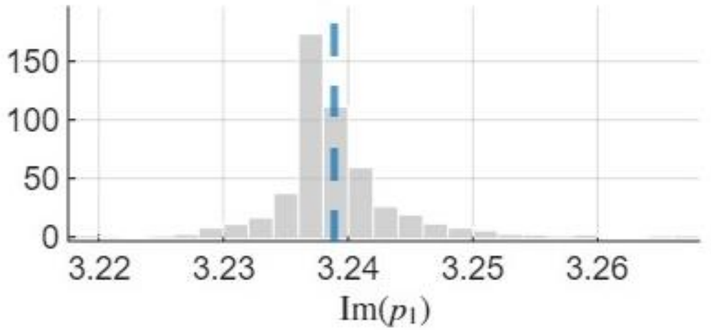
	Monte Carlo	Greyest
X_u	54.6471%	60.56%
X_q	11.2800%	19.40%
M_u	1.0494%	0.99%
M_q	0.4899%	0.54%
X_δ	3.8139%	6.36%
M_δ	0.0834%	0.13%

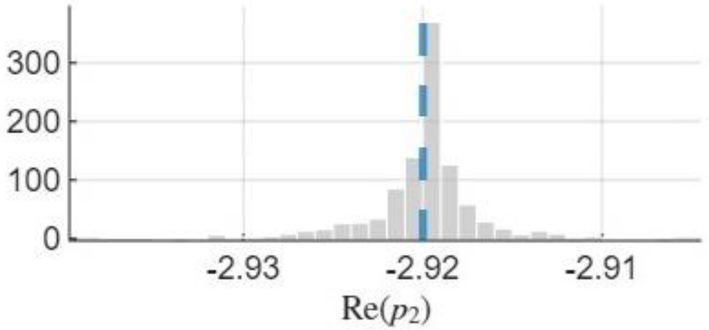
Monte Carlo analysis – Poles Distribution

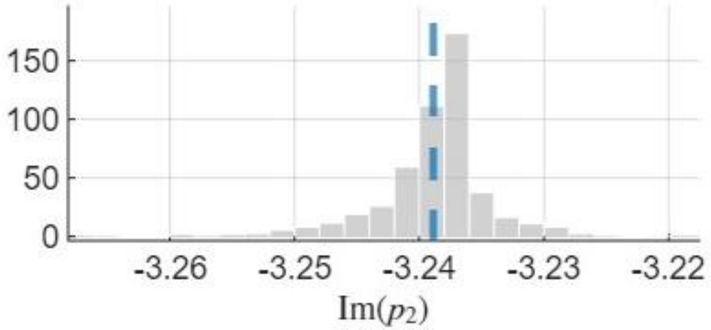
The System Has 3 Poles:

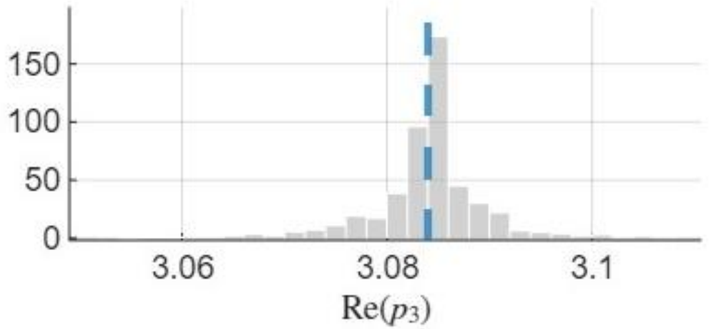
- A pair of complex conjugates
- A real pole (unstable in open loop)

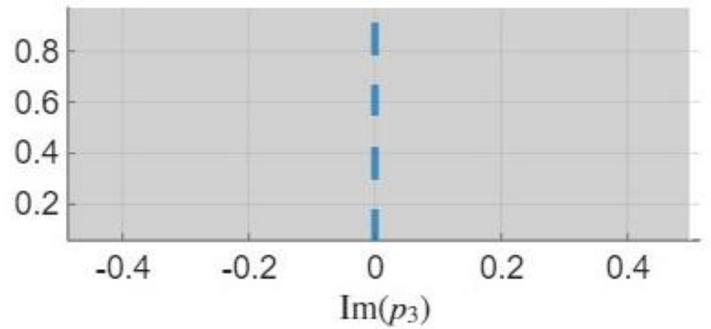




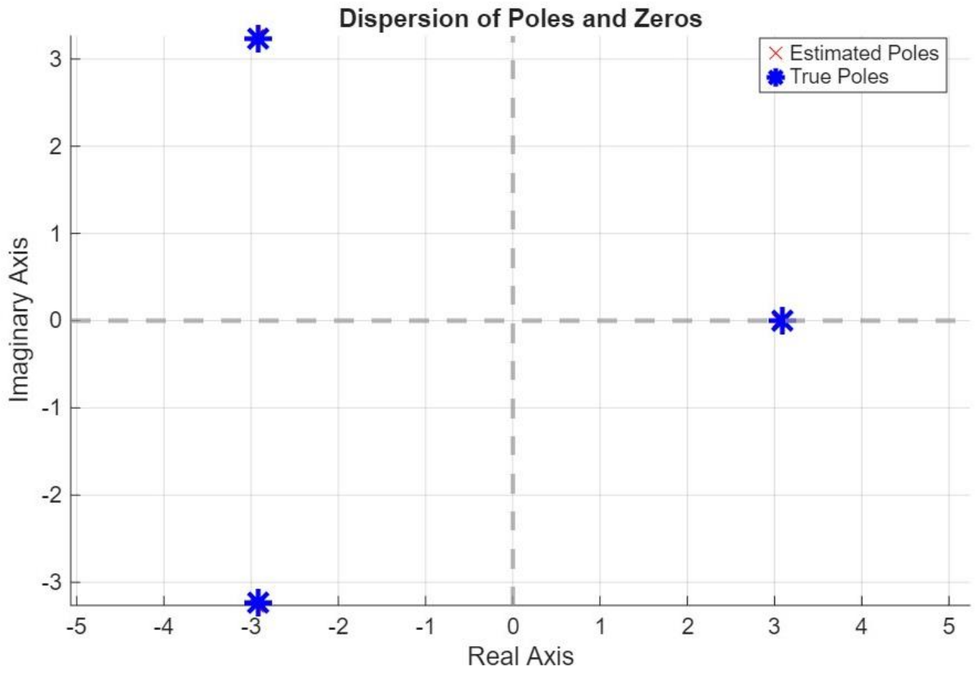


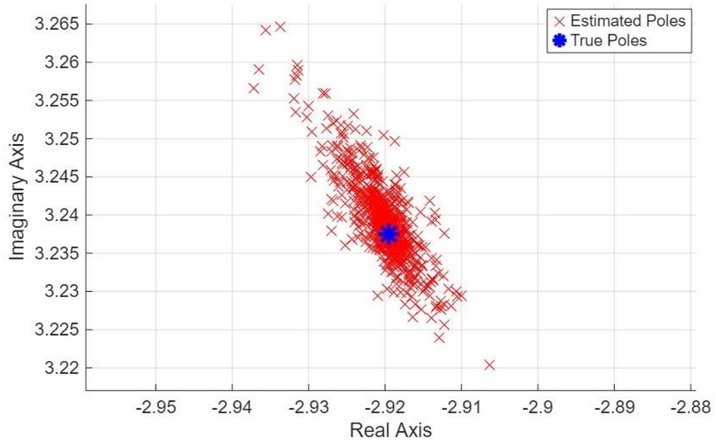


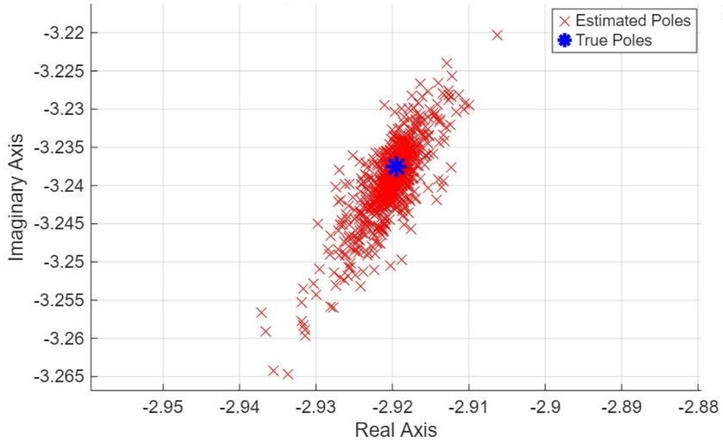


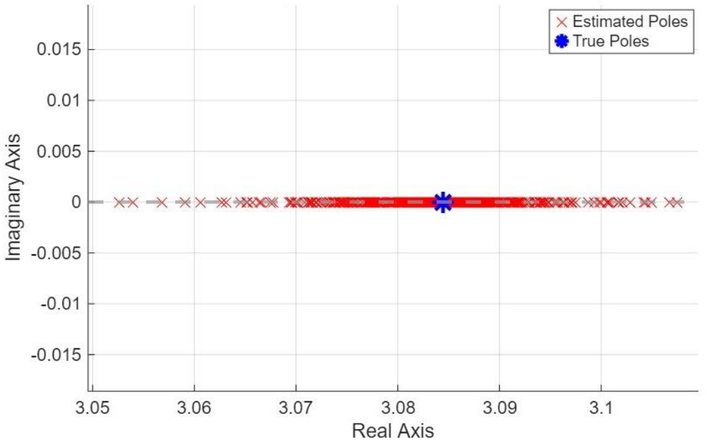


Monte Carlo analysis – Dispersion of poles





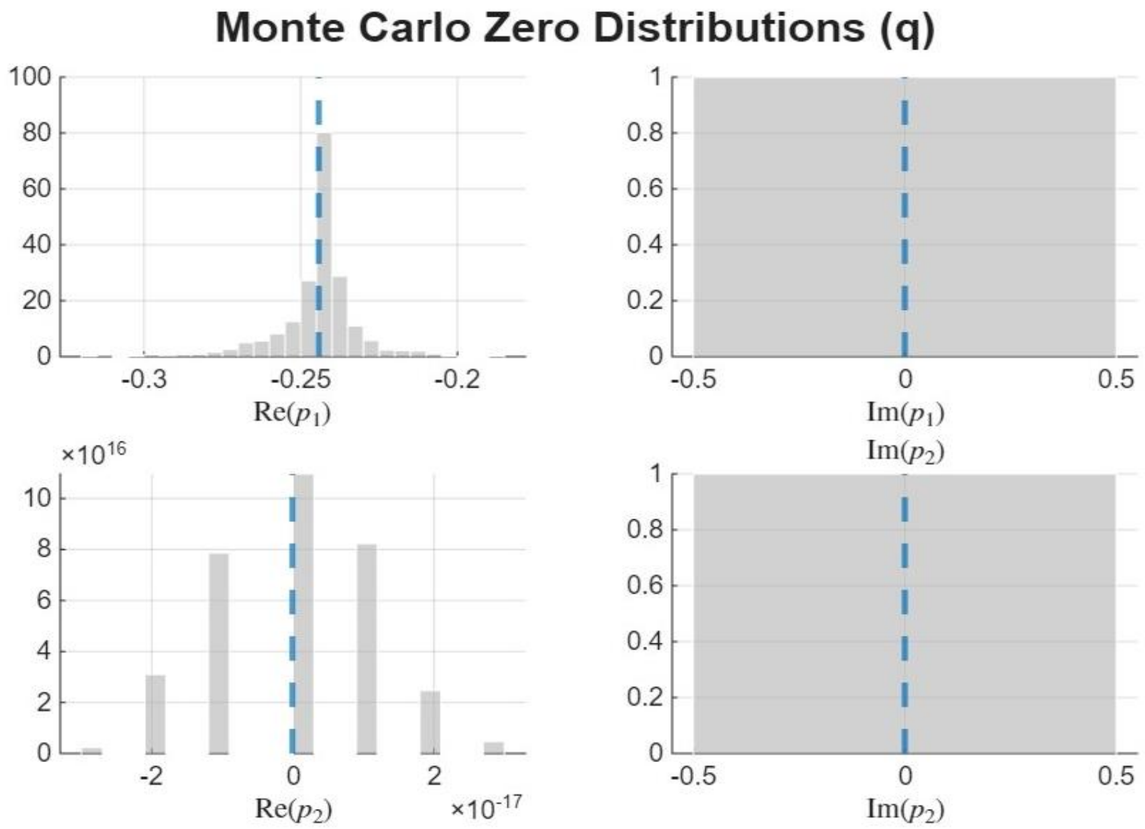


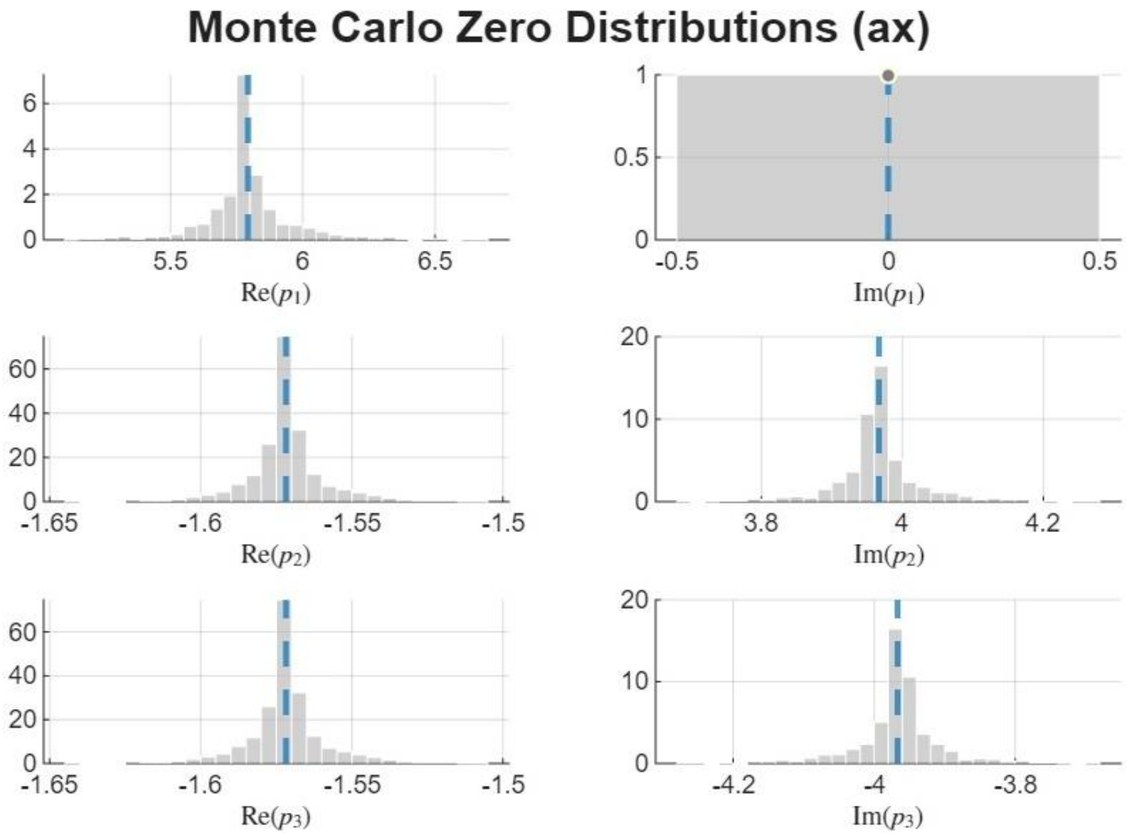


Monte Carlo analysis – Zeros of the System

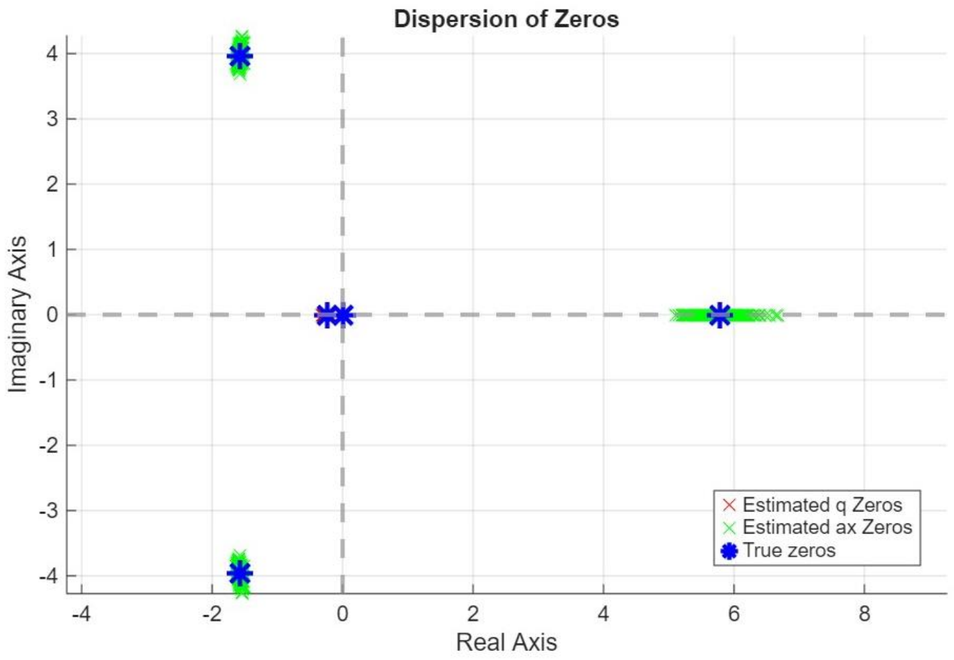
The System Has 3 Poles:

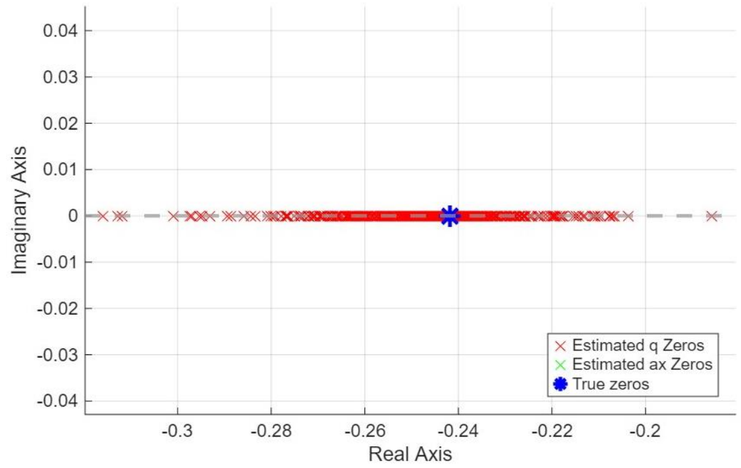
- 2 zeros for q
- 3 zeros for a_x

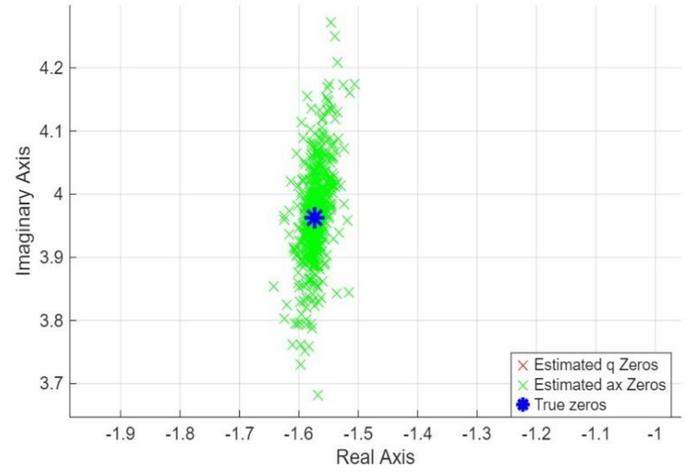


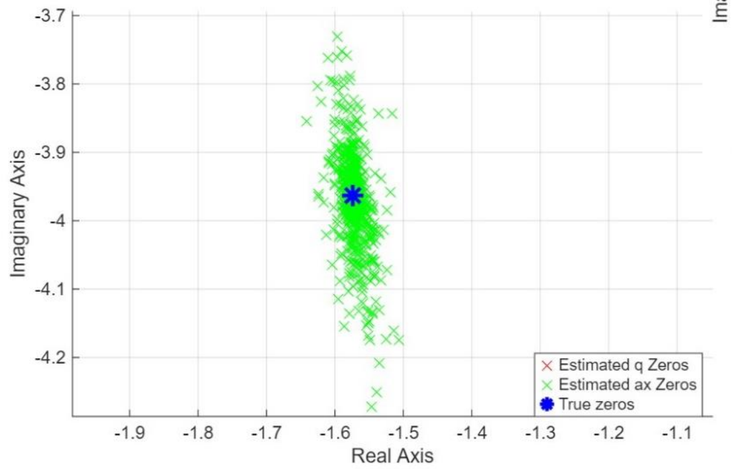


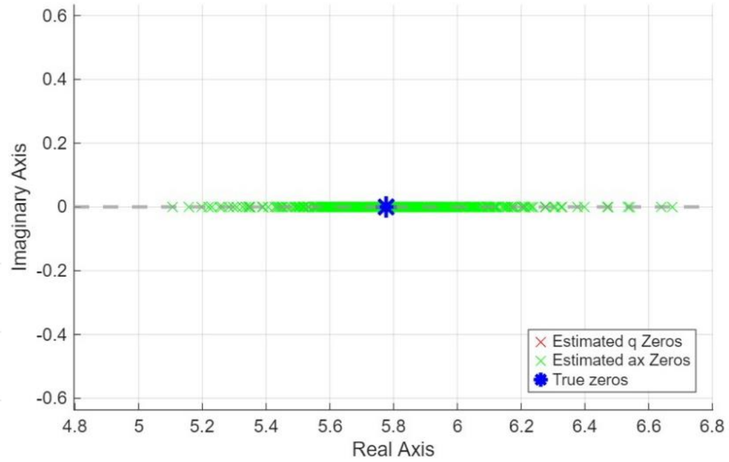
Monte Carlo analysis – Dispersion of zeros of the System



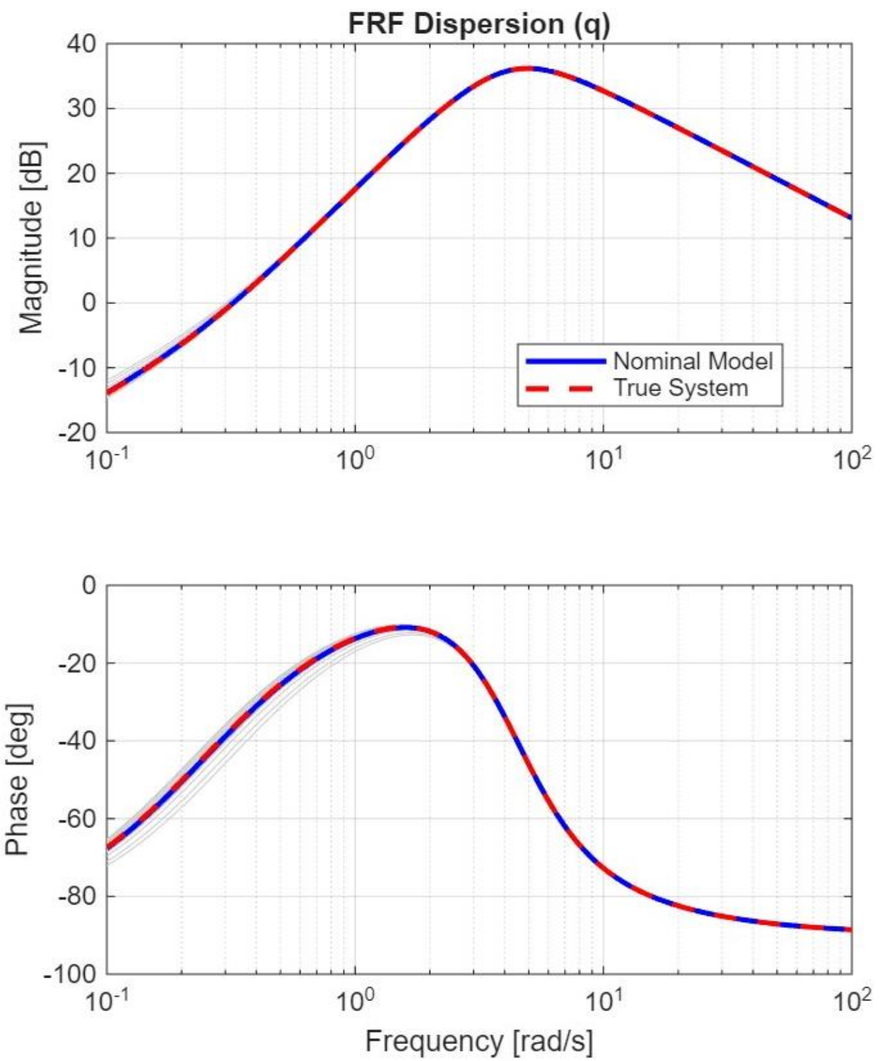


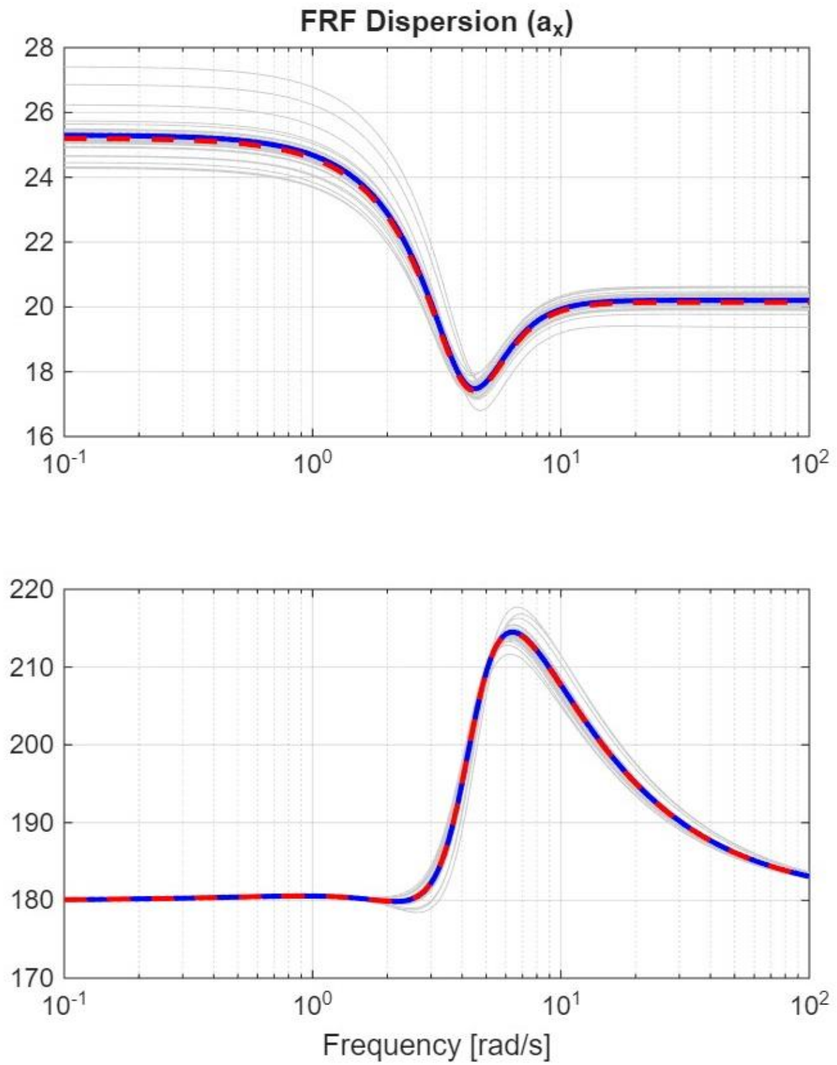






Monte Carlo analysis – FRF dispersions







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Thank you for your attention